Table 2. Measured CPU server cost: median [IQR]

Method	Broadcast		Full step	
	Time (ms)	ΔRSS (MiB)	Time (ms)	ΔRSS (MiB)
CIFAR10				
RBLA	2.03 [0.01]	2.24 [<0.01]	13.28 [0.43]	28.88 [0.21]
RBLA+	26.41 [0.31]	15.18 [0.10]	39.23 [0.72]	48.99 [0.11]
Dense SVD	20.80 [0.33]	13.23 [0.08]	34.21 [1.17]	47.00 [0.06]
MNLI				
RBLA	9.72 [0.47]	6.84 [2.53]	44.73 [0.36]	311.03 [11.25]
RBLA+	20.87 [0.84]	12.34 [0.08]	57.01 [0.53]	293.37 [30.29]
Dense SVD	186.48 [0.33]	8.54 [0.07]	212.64 [0.64]	313.19 [22.65]
QQP				
RBLA	9.64 [0.19]	5.87 [0.03]	44.66 [0.28]	311.04 [12.54]
RBLA+	20.44 [0.76]	10.68 [0.10]	56.47 [0.40]	277.90 [56.17]
Dense SVD	186.54 [0.28]	8.09 [0.04]	215.79 [1.04]	318.07 [26.59]

Median [IQR] across eight checkpoint inputs; each time is the median of 20 runs after five warmups. CPU FP32, one thread. ΔRSS is cold-pass peak incremental CPU RSS above resident inputs (1 ms sampling). Full step includes aggregation and all requested broadcasts; excludes I/O, transfers, model loading and evaluation. Dense SVD uses economy full SVD then truncation. Positive IQRs rounding to zero appear as <0.01.